GRADE 7 · CAMBRIDGE MATHS · 2ND TERM REVIEW

The Maths Study Guide

Everything on your 2nd Term worksheet, explained step by step by Pi the Robot: the method, the worksheet questions worked out line by line, the mistakes to avoid, and practice questions with answers.

CHAPTER 2
Expressions, formulae & equations

CHAPTER 4
Decimals

CHAPTER 6
Collecting data

CHAPTER 9
Sequences & functions

CHAPTER 10
Percentages

How to use this guide

There is one section for each of the 15 Maths lessons. Each section gives the **big idea**, the **method** in numbered steps, the **worksheet questions worked out** line by line, and **Try these** practice questions. The answers to every **Try these** question are in the **answer key** at the back. The **cheat sheet** has every rule on one page.

EXAM TIP
How to get the marks.

WATCH OUT!
A common mistake to avoid.

MEMORY TRICK
An easy way to remember.

TRY A NUMBER
Check a rule with a real number.

Tick off each lesson when you know it

- ☐ 2.1 Writing expressions
- ☐ 2.2 Substituting
- ☐ 2.3 Like terms
- ☐ 2.4 Expanding brackets
- ☐ 2.5 Solving equations
- ☐ 2.6 Inequalities
- ☐ 4.1 Ordering decimals
- ☐ 4.2 Mental methods
- ☐ 4.3 Multiplying decimals
- ☐ 4.4 Dividing decimals
- ☐ 6.1 Types of data
- ☐ 6.2 Mode, median, mean
- ☐ 9.1 BIDMAS & negatives
- ☐ 9.2 Sequences
- ☐ 9.3 nth term & machines
- ☐ 10.1 % ↔ fractions ↔ decimals
- ☐ 10.2 Percentages of amounts

2 Expressions, Formulae & Equations

Turning words into algebra, working out values, tidying expressions, expanding brackets, solving equations and inequalities.

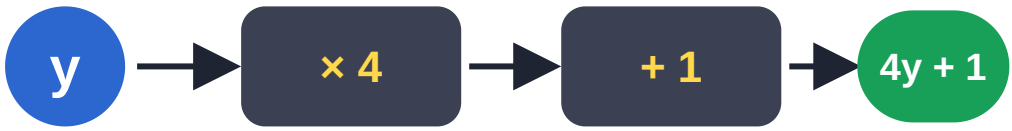
Big idea: A **letter** stands for a number we don't know yet. Algebra has rules for writing, tidying and solving with letters, and every rule works the same way it does with ordinary numbers. **If you're stuck, try a real number first.**

2.1 Writing expressions

An **expression** is a maths recipe with a letter in it, like **y + 3**: "whatever y is, add 3". There is no = sign in an expression.

The method

1. Find the **letter** (the unknown number).
2. Turn each word into a symbol: **more than / plus / increased by** → +; **less than / take away / decreased by** → −; **times / multiplied by / twice** → ×; **divided by / half** → ÷.
3. Do the steps **in the order the question says**.
4. Write it the short algebra way: no × sign, **number first** (3 × y = **3y**, p × q = **pq**), and ÷ as a fraction (y ÷ 2 = $\frac{y}{2}$).



A number machine: y goes in, is multiplied by 4, then 1 is added. Out comes 4y + 1.

Worksheet questions, worked out

WORKSHEET · EX 2.1 · Q3

The temperature on Tuesday was t°C. Write an expression for the temperature when it is: a) 2°C higher, b) twice as warm, c) half as warm as it was on Tuesday.

a) higher → add 2: **(t + 2)°C**

b) twice → × 2: **2t°C**

c) half → ÷ 2: **$\frac{t}{2}$°C** (or (t ÷ 2)°C)

WORKSHEET · EX 2.1 · Q6

Maliha thinks of a number 'y'. Write an expression for the number Maliha gets when she:

She…	Working	Answer
(i) multiplies the number by 3	y × 3	**3y**
(ii) divides the number by 2	y ÷ 2	**$\frac{y}{2}$ or y ÷ 2**
(iii) multiplies the number by 4, then adds 1	y × 4 = 4y, then + 1	**4y + 1**
(iv) multiplies the number by 2, then subtract 5	y × 2 = 2y, then − 5	**2y − 5**
(v) multiplies the number by 5, then subtract the result from 52	y × 5 = 5y, then 52 − 5y	**52 − 5y**
(vi) divides the number by 4, then add 3	y ÷ 4, then + 3	**$\frac{y}{4}$ + 3**

WORKSHEET · EX 2.1 · Q10

Write an expression for each of these.

a) x more than y → **x + y** (or y + x)
b) x less than y → start with y, take x away: **y − x**
c) m more than two times n → 2n, then add m: **m + 2n**
d) a less than 3 times b → 3b, then take a away: **3b − a**
e) p multiplied by q → **pq**
f) four times g multiplied by h → **4gh**

WATCH OUT!

"x less than y" is y − x, not x − y. With numbers: 3 less than 10 is 10 − 3 = 7. The words come in the *opposite* order to the expression. The same goes for "subtract the result **from 52**": 52 comes first.

TRY A NUMBER

Stuck? Pretend y = 5. "Multiply by 4, then add 1" gives 5 × 4 + 1 = 21. Now do exactly the same with the letter: 4y + 1.

✏️ **TRY THESE (ANSWERS AT THE BACK)**

a. Write an expression for 6 less than k.
b. Write an expression for: multiply n by 7, then add 2.
c. Write an expression for: subtract 3w from 20.

2.2 Substituting into expressions and formulae

Substitute means "swap in". When you know what the letter is worth, put that number in place of the letter, then work it out. A **formula** is a rule with an = sign that links letters, like **M = P − E**.

The method

1. Write the expression (or formula) down first.
2. Rewrite it with each letter swapped for its number. Put back the hidden signs: **2m** means 2 × m, **ab** means a × b, and a fraction line means ÷.
3. Work it out: **× and ÷ before + and −**. Work out the top of a fraction first.
4. For a formula, if the letter you want is not on its own, rearrange first (**M = P − E** gives **P = M + E**).



Swap each letter for its number, put the × back in, then work it out: 14 + 6 = 20.

Worksheet questions, worked out

WORKSHEET · EX 2.2 · Q1 AND Q2

Find the value of each expression.

Expression	Working	Answer	Expression	Working	Answer
Q1 a) x + 9, x = 13	13 + 9	**22**	Q2 b) x − 9, x = 20	20 − 9	**11**
Q1 b) $\frac{x}{5}$, x = 40	40 ÷ 5	**8**	Q2 c) f + g, f = 7, g = 4	7 + 4	**11**
Q1 c) 10 − x, x = 3	10 − 3	**7**	Q2 d) m − n, m = 100, n = 25	100 − 25	**75**
Q1 d) 2m + n, m = 7, n = 6	2 × 7 + 6 = 14 + 6	**20**	Q2 e) 3k, k = 5	3 × 5	**15**

Q1 e) 7x, x = 5	7 × 5	**35**	Q2 f) p + 2q, p = 5, q = 3	5 + 2 × 3 = 5 + 6	**11**
Q1 f) ab − c, a = 9, b = 8, c = 32	9 × 8 − 32 = 72 − 32	**40**	Q2 g) 2h + 3t, h = 8, t = 5	2 × 8 + 3 × 5 = 16 + 15	**31**
Q2 a) a + 5, a = 3	3 + 5	**8**	Q2 h) $\frac{y}{4}$, y = 32	32 ÷ 4	**8**
Q2 i) $\frac{30}{c}$ − 2, c = 6	30 ÷ 6 − 2 = 5 − 2	**3**	Q2 j) $\frac{x + y}{2}$, x = 19, y = 11	(19 + 11) ÷ 2 = 30 ÷ 2	**15**

WORKSHEET · CH 9 GETTING STARTED · Q4

Work out the value of each expression. a) n + 4 when n = 5 b) 6n when n = 3 c) n − 9 when n = 20 d) n − 10 when n = 8

a) 5 + 4 = **9** b) 6 × 3 = **18** c) 20 − 9 = **11**

d) 8 − 10 = **−2** (start at 8 and go down 10: past 0 to −2)

WORKSHEET · EX 2.2 · Q8

Jan calculates how much money he has left each week, using the formula M = P − E, where M is the money he has left over, P is the money he is paid for working and E is his expenses. a) Find the value of M when: (i) P = $225 and E = $72, (ii) P = $178 and E = $36. b) Work out the value of P when M = $160, E = $45.

a) (i) M = 225 − 72 = **$153** (ii) M = 178 − 36 = **$142**

b) P is the pay, so P = M + E = 160 + 45 = **$205** (check: 205 − 45 = 160 ✓)

WATCH OUT!

2m with m = 7 is 2 × 7 = 14, not 27. The letter next to a number means multiply. And in **p + 2q**, do the × first: 5 + 2 × 3 = 5 + 6 = 11, not 7 × 3.

TIP

Write the expression, then the line with the numbers in, then the answer. Showing the middle line earns marks even if you slip later.

✏️ **TRY THESE (ANSWERS AT THE BACK)**

a. Find the value of 4p − 3 when p = 6.

b. Find the value of $\frac{a + b}{3}$ when a = 14 and b = 13.

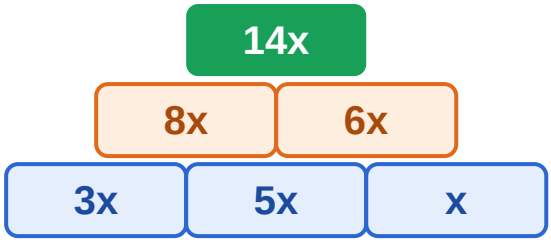
c. Using M = P − E, work out the value of P when M = $90 and E = $38.

2.3 Simplifying and like terms

Like terms have exactly the same letter part: **5x** and **2x** are like terms; **5x** and **2y** are not. To **simplify**, add or take away the numbers in front of like terms. Think of x as "apples": 5 apples + 4 apples = 9 apples.

The method

1. Find the like terms. Each term keeps the sign **in front of it** (− 3p goes with the minus).
2. Move the like terms next to each other, signs and all.
3. Add or take away the numbers in front. A letter on its own counts as 1 (**x** = **1x**), and you write **1x** as just **x**.
4. Numbers with no letter are like terms too: collect them on their own.



An algebra pyramid: each block is the two blocks under it added. 3x + 5x = 8x, 5x + x = 6x, 8x + 6x = 14x.

Worksheet questions, worked out

WORKSHEET · EX 2.3 · Q2

Match each expression, a to f, to its correctly simplified expression, i to vi. (i) 7x (ii) x (iii) 4x (iv) 9x (v) 2x (vi) 3x

a) x + x = 1x + 1x = 2x → **(v)** b) 5x + 4x = 9x → **(iv)** c) x + 6x = 7x → **(i)**
d) 5x − 2x = 3x → **(vi)** e) 2x − x = 1x = x → **(ii)** f) 8x − 4x = 4x → **(iii)**

WORKSHEET · EX 2.3 · Q3

Simplify each of these expressions.

Question	Answer	Question	Answer	Question	Answer
a) x + x + x + x + x	**5x**	e) 8g + 5g + g	**14g**	i) $9b^2 - 5b^2$	**$4b^2$**
b) 2y + 4y	**6y**	f) 9p + p + 6p	**16p**	j) 6f + 2f − 3f = 8f − 3f	**5f**
c) 5d + 3d	**8d**	g) 7w − 4w	**3w**	k) 9j + j − 7j = 10j − 7j	**3j**
d) 6t + 3t + 4t	**13t**	h) 8n − n	**7n**	l) $8k^3 - 5k^3 - 2k^3 = 3k^3 - 2k^3$	**k^3**

WORKSHEET · EX 2.3 · Q4 AND Q5

Q4: In an algebra pyramid, you find the expression in each block by adding the expressions in the two blocks below it. Complete the pyramids. Q5: Complete the workings to simplify these expressions.

Q4 a) bottom 3x, 5x, x → middle 8x, 6x → top **14x** (see the picture) b) bottom 7p, p, 3p → middle **8p**, **4p** → top **12p**

Q5 a) 2x + 8x + 6y + 9y = **10x + 15y**

b) 9d − 7d + 3h − h = **2d + 2h**

c) 2g + 8 + 3g − 5 = 2g + 3g + 8 − 5 = **5g + 3**

d) 8p + 12t − 3p + t = 8p − 3p + 12t + t = **5p + 13t**

e) 3a + 4b + 5c − 2b − 8c = 3a + 4b − 2b + 5c − 8c = **3a + 2b − 3c** (5c − 8c = −3c)

WATCH OUT!

Only like terms join. 10x + 15y cannot be made any shorter, and **5g + 3** is not 8g. Also, **b^2** and **b** are *not* like terms. Keep the power the same: $9b^2 - 5b^2 = 4b^2$.

MEMORY TRICK

Carry the sign with the term. Circle each term together with the sign in front of it, then shuffle the circles so like terms sit together.

✏️ **TRY THESE (ANSWERS AT THE BACK)**

a. Simplify 7m + 2m − 4m.

b. Simplify 6a + 5 + 2a − 3.

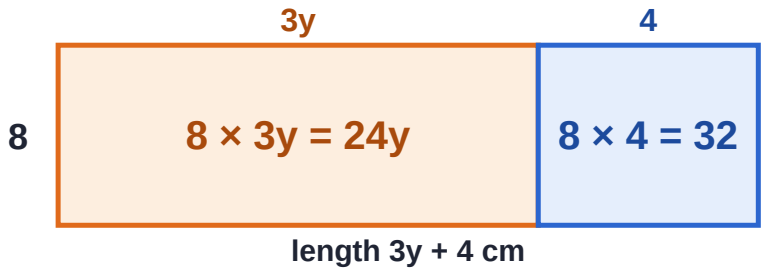
c. An algebra pyramid has 2y, 4y, 3y on the bottom row. Find the two middle blocks and the top block.

2.4 Expanding brackets

Expand (or "multiply out") means get rid of the brackets. The number outside multiplies **every** term inside: **3(y + 6)** = 3 × y + 3 × 6 = **3y + 18**.

The method

1. Multiply the number outside by the **first** term inside.
2. Multiply it by the **next** term, and keep the sign in front of that term (+ stays +, − stays −).
3. Do the same for every term inside: there may be three.
4. Check: one term out for each term in.



Area = 8(3y + 4). Split the rectangle in two: 8 × 3y = 24y and 8 × 4 = 32, so the area is 24y + 32.

Worksheet questions, worked out

WORKSHEET · EX 2.4 · Q3

Expand the brackets.

Question	Answer	Question	Answer	Question	Answer
a) 3(y + 6)	**3y + 18**	e) 6(d − 9)	**6d − 54**	i) 9(3 + i)	**27 + 9i**
b) 4(w + 2)	**4w + 8**	f) 2(e − 8)	**2e − 16**	j) 6(2 − x)	**12 − 6x**
c) 5(z + 5)	**5z + 25**	g) 6(2 + f)	**12 + 6f**	k) 2(1 − y)	**2 − 2y**
d) 3(b − 1)	**3b − 3**	h) 2(l + g)	**2l + 2g**	l) 5(7 − p)	**35 − 5p**

WORKSHEET · EX 2.4 · Q5

Multiply out the brackets.

Question	Working	Answer	Question	Working	Answer
a) 3(2x + 1)	3 × 2x, 3 × 1	**6x + 3**	g) 6(5d − 1)	6 × 5d, 6 × 1	**30d − 6**
b) 4(3y + 5)	4 × 3y, 4 × 5	**12y + 20**	h) 8(3e − 6 + 2f)	8 × 3e, 8 × 6, 8 × 2f	**24e − 48 + 16f**
c) 5(2w + 3)	5 × 2w, 5 × 3	**10w + 15**	i) 3(a + 2f)	3 × a, 3 × 2f	**3a + 6f**
d) 6(4z + 7v + 9)	6 × 4z, 6 × 7v, 6 × 9	**24z + 42v + 54**	j) 5(3b + 4g)	5 × 3b, 5 × 4g	**15b + 20g**
e) 2(3b − 4)	2 × 3b, 2 × 4 (keep the −)	**6b − 8**	k) 7(6c − 7h)	7 × 6c, 7 × 7h	**42c − 49h**
f) 4(2c − 3)	4 × 2c, 4 × 3	**8c − 12**	l) 9(5 + 3h − 4i)	9 × 5, 9 × 3h, 9 × 4i	**45 + 27h − 36i**

WORKSHEET · EX 2.4 · Q9

The diagram shows a rectangle. The width of the rectangle is 8 cm. The length of the rectangle is 3y + 4 cm. Write an expression, in its simplest form, for the: a) area of the rectangle, b) perimeter of the rectangle.

> a) Area = width × length = 8(3y + 4) = **24y + 32 cm²**
>
> b) Perimeter = 2(width + length) = 2(8 + 3y + 4) = 2(3y + 12) = **6y + 24 cm** (tidy inside the bracket first: 8 + 4 = 12)

WATCH OUT!

Multiply every term, and keep the minus. 2(3b − 4) is 6b **−** 8, not 6b + 8, and not 6b − 4. The 2 multiplies the 4 too.

TRY A NUMBER

Check with b = 1: 2(3 × 1 − 4) = 2 × (−1) = −2, and 6 × 1 − 8 = −2 too. ✓ Same answer, so the expansion is right.

✏️ **TRY THESE (ANSWERS AT THE BACK)**

a. Expand 5(k − 3).

b. Multiply out 4(2x + 7).

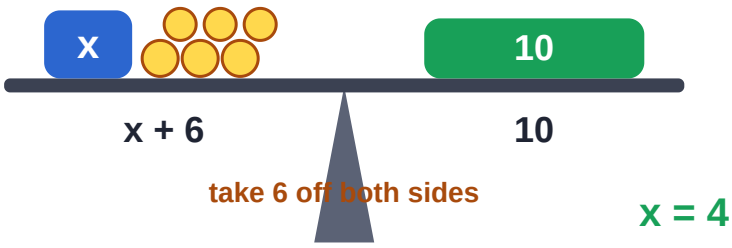
c. Expand 3(6 − 2y).

2.5 Solving equations

An **equation** has an = sign: both sides are worth the same, like a balanced scale. **Solving** means finding the number the letter stands for. Whatever you do to one side, do to the other side, and it stays balanced.

The method

1. Look at what is being done to x.
2. Do the **opposite** to **both sides**: + ↔ −, × ↔ ÷. (x + 6: take away 6. x − 6: add 6. 2x: divide by 2.)
3. Keep going until x is on its own: **x = …**
4. **Check:** put your answer back into the first equation. Both sides must match.



x + 6 = 10 on a balance. Take 6 from both pans and it still balances: x = 4.

Worksheet questions, worked out

WORKSHEET · EX 2.5 · Q1

Copy and complete to solve these equations. Check your answers are correct. a) x + 6 = 10 b) x − 6 = 10 c) 2x = 10

a) x + 6 − 6 = 10 − 6, so **x = 4** check: 4 + 6 = 10 ✓

b) x − 6 + 6 = 10 + 6, so **x = 16** check: 16 − 6 = 10 ✓

c) 2x ÷ 2 = 10 ÷ 2, so **x = 5** check: 2 × 5 = 10 ✓

WORKSHEET · EX 2.5 · Q2

Solve each of these equations and check your answers.

Equation	Do this to both sides	Answer	Equation	Do this to both sides	Answer
a) x + 4 = 11	− 4: 11 − 4	**x = 7**	g) x − 12 = 14	+ 12: 14 + 12	**x = 26**
b) x + 3 = 6	− 3: 6 − 3	**x = 3**	h) x − 18 = 30	+ 18: 30 + 18	**x = 48**
c) 2 + x = 15	− 2: 15 − 2	**x = 13**	i) 3x = 12	÷ 3: 12 ÷ 3	**x = 4**
d) 7 + x = 19	− 7: 19 − 7	**x = 12**	j) 5x = 30	÷ 5: 30 ÷ 5	**x = 6**
e) x − 4 = 9	+ 4: 9 + 4	**x = 13**	k) 7x = 70	÷ 7: 70 ÷ 7	**x = 10**
f) x − 2 = 8	+ 2: 8 + 2	**x = 10**	l) 12x = 72	÷ 12: 72 ÷ 12	**x = 6**

Check each one by putting x back in, e.g. h) 48 − 18 = 30 ✓, l) 12 × 6 = 72 ✓.

WATCH OUT!

Do the opposite, not the same. For x − 4 = 9, add 4 (x = 13), don't take it away (x = 5 is wrong: 5 − 4 = 1, not 9). For 3x = 12, divide by 3 (x = 4); don't take away 3.

TIP

The check takes ten seconds and catches nearly every mistake. Always write it.

✏️ **TRY THESE (ANSWERS AT THE BACK)**

a. Solve x + 9 = 23.

b. Solve x − 7 = 15.

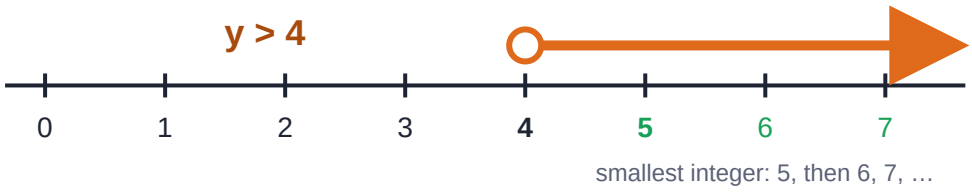
c. Solve 8x = 56.

2.6 Inequalities

An **inequality** says one side is bigger than the other. **<** means "is less than" and **>** means "is greater than". **x > 4** means x could be any number bigger than 4, but not 4 itself.

The method

1. Read the sign from left to right: **x < 10** is "x is less than 10"; **x > 10** is "x is greater than 10".
2. The **open (wide) end** of the sign faces the bigger number.
3. On a number line: an **open circle** on the number (it is not allowed), and an arrow towards the numbers that work.
4. An **integer** is a whole number (it can be negative). For **y > 4**, the smallest integer is the next whole number **up**: 5.



y > 4: an open circle at 4 (4 is not allowed) and an arrow to the right. The integers that work are 5, 6, 7, …

Worksheet questions, worked out

WORKSHEET · EX 2.6 · Q1 AND Q2

Q1: Write in words what each of these inequalities means. Q2: Write these statements as inequalities.

Q1 a) x < 10: **x is less than 10**

b) x > 10: **x is greater than 10**

c) x < −4: **x is less than −4**

d) x > −4: **x is greater than −4**

Q2 a) y is greater than 8: **y > 8**

b) n is greater than −1: **n > −1**

c) p is less than zero: **p < 0**

d) q is less than −2: **q < −2**

WORKSHEET · EX 2.6 · Q6

For each of these inequalities: a) y > 4 b) y > −7 c) y > 2.5 (i) Write down the smallest integer that y could be. (ii) Write a list of the integer values that y could be.

Inequality	(i) Smallest integer	(ii) Integer values	Why
a) y > 4	**5**	**5, 6, 7, …**	4 is not allowed; the next whole number up is 5
b) y > −7	**−6**	**−6, −5, −4, …**	going up from −7 means towards 0: −6 is bigger than −7
c) y > 2.5	**3**	**3, 4, 5, …**	2.5 is not whole; the next whole number up is 3

WATCH OUT!

Negative numbers go the other way. −6 is *greater* than −7 (it is nearer 0, further right on the number line). So the smallest integer for $y > -7$ is −6, not −8.

MEMORY TRICK

The crocodile eats the bigger number. The open mouth of < or > always faces the bigger side: $3 < 8$, $8 > 3$.

✏️ **TRY THESE (ANSWERS AT THE BACK)**

a. Write in words what $m < 3$ means.

b. Write as an inequality: k is greater than −3.

c. For $y > 6.5$, write the smallest integer y could be, and a list of the integer values y could be.

4 Decimals

Ordering and comparing decimals, quick mental methods, multiplying decimals, dividing decimals and sharing money.

Big idea: Every digit in a decimal has a **place value**: ones, then **tenths**, **hundredths**, **thousandths**. Line the digits up by place value and decimals work just like whole numbers. **Always estimate first**, so you can spot a decimal point in the wrong place.

4.1 Ordering and comparing decimals

To compare two decimals, compare them **place by place**, starting from the left. The first place where the digits differ decides which number is bigger.

The method

1. Compare the **whole-number parts** first. If one is bigger, that number is bigger.
2. If they are the same, line up the decimal points and **pad with zeros** so both have the same number of decimal places (2.19 → 2.190).
3. Compare the **tenths**, then the **hundredths**, then the **thousandths**, until two digits are different.
4. Write the sign: **<** means "is smaller than", **>** means "is bigger than". **=** means equal, **≠** means "not equal".

	Ones		Tenths	Hundredths	Thousandths	
	2	.	1	9	0	← padded zero
	2	.	2	0	5	1 < 2, so 2.19 < 2.205

Line up the place values and pad 2.19 to 2.190. The ones are equal, so look at the tenths: 1 tenth is less than 2 tenths.

Worksheet questions, worked out

WORKSHEET · EX 4.1 · Q1

Write down which is the smallest decimal number.

Numbers	Why	Smallest
a) 13.5, 9.99	9 ones < 13 ones	**9.99**
b) 4.32, 3.67	3 ones < 4 ones	**3.67**
c) 12.56, 21.652	12 < 21	**12.56**
d) 127.06, 246.9	127 < 246	**127.06**
e) 0.67, 0.72	tenths: 6 < 7	**0.67**
f) 3.4, 3.21	tenths: 2 < 4 (3.21 vs 3.40)	**3.21**
g) 18.54, 18.45	tenths: 4 < 5	**18.45**
h) 0.05, 0.043	0.050 vs 0.043: hundredths 4 < 5	**0.043**
i) 0.09, 0.1	0.09 vs 0.10: tenths 0 < 1	**0.09**

WORKSHEET · EX 4.1 · Q4

Greg uses the inequalities < and > to show that one number is smaller than or is larger than another number. Write the correct sign, < or >.

Question	First place that differs	Answer
a) 6.03 ☐ 6.24	tenths: 0 < 2	**6.03 < 6.24**
b) 9.35 ☐ 9.41	tenths: 3 < 4	**9.35 < 9.41**
c) 0.49 ☐ 0.51	tenths: 4 < 5	**0.49 < 0.51**
d) 18.05 ☐ 18.02	hundredths: 5 > 2	**18.05 > 18.02**
e) 9.2 ☐ 9.01	tenths: 2 > 0	**9.2 > 9.01**
f) 2.19 ☐ 2.205	tenths: 1 < 2	**2.19 < 2.205**
g) 0.072 ☐ 0.06	hundredths: 7 > 6	**0.072 > 0.06**
h) 29.882 ☐ 29.82	hundredths: 8 > 2	**29.882 > 29.82**

WORKSHEET · EX 4.1 · Q5

Write the correct sign = or ≠

a) 4.2 and 4.20: a zero on the end changes nothing → **4.2 = 4.20**
b) 3.75 and 3.57: same digits, different places → **3.75 ≠ 3.57**
c) 0.340 and 0.304: 4 hundredths vs 0 hundredths → **0.340 ≠ 0.304**
d) **9.58 = 9.580** (end zero) e) **128.00 = 128** (no tenths, no hundredths)
f) 0.0034 and 0.034: the zero is **inside**, so it moves the digits → **0.0034 ≠ 0.034**

WATCH OUT!

The longer decimal is not always bigger. 2.205 has more digits than 2.19, but that is not why it is bigger; 3.21 has more digits than 3.4 and is *smaller*. Always compare place by place. Zeros on the **end** after the point change nothing (4.2 = 4.20); zeros **straight after the point** do (0.034 ≠ 0.0034).

MEMORY TRICK

The sign is a hungry crocodile: its mouth opens towards the **bigger** number. 9.2 > 9.01.

✏️ **TRY THESE (ANSWERS AT THE BACK)**

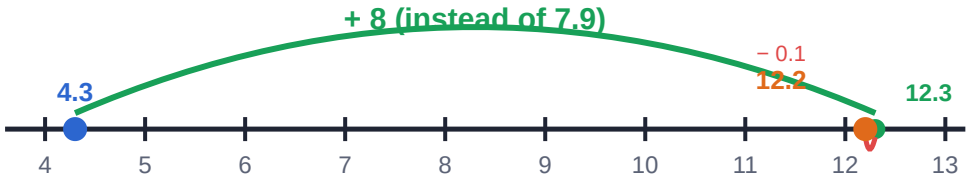
a. Which is the smallest: 0.8 or 0.75?
b. Write < or >: 5.306 ☐ 5.31
c. Write in order, smallest first: 3.4, 3.04, 3.399, 3.41

4.2 Mental methods with decimals

Many decimal sums can be done in your head. Split numbers into **wholes and decimals**, or round one number to a **whole number** and then adjust. For harder ones, use a written column method.

The method

1. **Split:** add (or take away) the wholes, then the decimals, then put them together. 12.7 + 4.5 → 16 + 1.2 = 17.2.
2. **Zara's method** for subtracting: wholes first, then decimals. The decimal part may go below zero: 7.4 − 2.6 → 5 and −0.2 → 4.8.
3. **Round and adjust:** round one number to a whole, work it out, then undo the rounding. Added too much? Take it back off. Took away too much? Add it back.
4. **Times by a decimal:** 0.3 is 3 tenths, so 4 × 0.3 = 12 tenths = 1.2.
5. **Written method:** line up the decimal points, pad with zeros, then add or subtract in columns.



4.3 + 7.9: add 8 instead (easy!) to get 12.3. That was 0.1 too much, so take 0.1 back off: 12.2.

Worksheet questions, worked out

WORKSHEET · EX 4.2 · Q1

Use a mental method to work out the answer to the following.

a) 3.5 + 4.2 → 3 + 4 = 7 and 0.5 + 0.2 = 0.7 → **7.7**

b) 12.7 + 4.5 → 12 + 4 = 16 and 0.7 + 0.5 = 1.2 → 16 + 1.2 = **17.2**

c) 4.9 − 1.5 → 4 − 1 = 3 and 0.9 − 0.5 = 0.4 → **3.4**

d) 14.6 − 6.6 → both end in .6, so the decimals cancel: 14 − 6 = **8**

WORKSHEET · EX 4.2 · Q4

Use Zara's method to mentally work out

Question	Wholes	Decimals	Put together
a) 7.4 − 2.6	7 − 2 = 5	0.4 − 0.6 = −0.2	**5 − 0.2 = 4.8**
b) 8.3 − 2.9	8 − 2 = 6	0.3 − 0.9 = −0.6	**6 − 0.6 = 5.4**
c) 12.5 − 9.8	12 − 9 = 3	0.5 − 0.8 = −0.3	**3 − 0.3 = 2.7**
d) 15.1 − 5.7	15 − 5 = 10	0.1 − 0.7 = −0.6	**10 − 0.6 = 9.4**

WORKSHEET · EX 4.2 · Q6

Use the method of rounding one of the numbers to a whole number.

Question	Round	Work out	Adjust
a) 4.3 + 7.9	7.9 → 8 (+0.1)	4.3 + 8 = 12.3	**− 0.1 → 12.2**
b) 8.9 + 9.6	8.9 → 9 (+0.1)	9 + 9.6 = 18.6	**− 0.1 → 18.5**
c) 22.8 + 3.3	22.8 → 23 (+0.2)	23 + 3.3 = 26.3	**− 0.2 → 26.1**
d) 5.4 − 1.9	1.9 → 2 (took 0.1 too much)	5.4 − 2 = 3.4	**+ 0.1 → 3.5**
e) 14.9 − 4.4	14.9 → 15 (+0.1)	15 − 4.4 = 10.6	**− 0.1 → 10.5**
f) 21.1 − 6.7	6.7 → 7 (took 0.3 too much)	21.1 − 7 = 14.1	**+ 0.3 → 14.4**

WORKSHEET · EX 4.2 · Q7 AND Q15

Use a written method to work out the following.

Q7 a) 25.81 + 8.40 (pad with a zero) → **34.21** b) 8.76 − 4.10 → **4.66** c) 38.91 − 9.78 → **29.13**

Q15 a) −5.43 − 9.57: both take you further below zero, so add 5.43 + 9.57 = 15.00 and keep the minus → **−15**

Q15 b) −8.12 − 15.4: add 8.12 + 15.40 = 23.52, keep the minus → **−23.52**

Q15 d) −8.12 − −15.4: minus a minus is plus, so −8.12 + 15.4 = 15.40 − 8.12 → **7.28**

WORKSHEET · EX 4.3 · Q1 AND Q2

Use a mental method to work out the following.

Q1 a) 0.1 × 8 = 8 tenths = **0.8** b) 0.5 × 5 = 25 tenths = **2.5** c) 0.9 × 2 = 18 tenths = **1.8**

Q2 a) 6 × 0.02 = 12 hundredths = **0.12** b) 4 × 0.3 = 12 tenths = **1.2**

Q2 c) 3 × 0.004 = 12 thousandths = **0.012** d) 120 × 0.1 = 120 tenths = **12** (× 0.1 is the same as ÷ 10)

WATCH OUT!

When you round and adjust, think about **which way** to adjust. In 5.4 − 1.9 you took away 2, which is 0.1 **too much**, so you must **add** 0.1 back: 3.5, not 3.3. And in a written method, always line up the **decimal points**, not the last digits.

EXAM TIP

Check a subtraction by adding back: 2.7 + 9.8 = 12.5 ✔. If it doesn't come back to the starting number, look again.

✏️ **TRY THESE (ANSWERS AT THE BACK)**

a. Use Zara's method: 9.2 − 3.7

b. Round one number to a whole number: 5.6 + 2.9

c. Work out mentally: 7 × 0.03

4.3 Multiplying decimals

To multiply decimals, forget the decimal points for a moment, multiply whole numbers, then put the point back. An **estimate** tells you if the point is in the right place.

The method

1. **Estimate** by rounding: 3.2 × 52 ≈ 3 × 50 = 150.
2. **Ignore the points** and multiply the whole numbers: 32 × 52 = 1664.
3. **Count the decimal places** in the question (digits after the points): 3.2 has 1.
4. Give the answer the **same number of decimal places**: 166.4. Check it is close to your estimate.
5. Shortcuts: **0.6 × 410** = 6 tenths of 410 = 6 × 41; **× 9** = × 10, then take one lot away.



1 decimal place — ignore the point, multiply — put back 1 decimal place

estimate 150 ✔

Ignore the point, multiply, then count the decimal places and put them back. The estimate (3 × 50 = 150) shows 166.4 is sensible.

Worksheet questions, worked out

WORKSHEET · EX 4.3 · Q8

Work out these multiplications. Show how to check your answer.

Question	Ignore the points	Decimal places	Answer	Check
a) 3.2 × 52	32 × 52 = 1664	1	**166.4**	3 × 50 = 150
b) 8.1 × 384	81 × 384 = 31104	1	**3110.4**	8 × 400 = 3200
c) 0.78 × 41	78 × 41 = 3198	2	**31.98**	0.8 × 40 = 32

WORKSHEET · EX 4.5 · Q2

Work out the answer of the following.

a) 0.6 × 410 = 6 × 41 = 6 × 40 + 6 × 1 = 240 + 6 = **246** (0.6 = 6 tenths; a tenth of 410 is 41)

b) 0.9 × 320 = 9 × 32 = 9 × 30 + 9 × 2 = 270 + 18 = **288**

c) 0.02 × 3200 = 2 × 32 = **64** (0.02 = 2 hundredths; a hundredth of 3200 is 32)

WORKSHEET · EX 4.5 · Q6 AND Q8

Work out the answer to the following.

Q6 a) 6.8 × 9 = 6.8 × 10 − 6.8 = 68 − 6.8 = **61.2**

Q6 b) 4.7 × 9 = 47 − 4.7 = **42.3** c) 12.6 × 9 = 126 − 12.6 = **113.4**

Q8 a) 4.2 × 6: 42 × 6 = 252 → **25.2** b) 7.8 × 5: 78 × 5 = 390 → 39.0 = **39** c) 6.3 × 8: 63 × 8 = 504 → **50.4**

WATCH OUT!

A misplaced decimal point makes the answer 10 times too big or too small. 3.2 × 52 is not 1664 or 16.64: the estimate 150 says 166.4. Count the decimal places in the **question**, not in your working.

ESTIMATE FIRST

Round each number to one easy digit before you start: 0.78 × 41 ≈ 0.8 × 40 = 32. If your answer is nowhere near 32, the point is in the wrong place.

✏️ **TRY THESE (ANSWERS AT THE BACK)**

a. Work out 2.6 × 34. Show how to check your answer.

b. Work out 0.7 × 520.

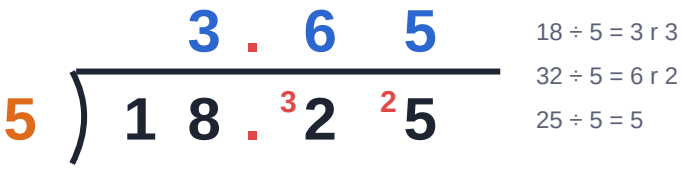
c. Work out 5.3 × 9 (use 9 = 10 − 1).

4.4 Dividing decimals and money problems

To divide a decimal by a whole number, use **short division** (the "bus stop"). The decimal point in the answer goes **straight above** the point in the number. Sharing money equally is just division.

The method

1. **Estimate** first: 18.25 ÷ 5 ≈ 20 ÷ 5 = 4.
2. Write the bus stop. Put the **decimal point in the answer** straight above the one inside.
3. Divide digit by digit from the left. Carry each **remainder** to the next digit.
4. If the divisor has two digits (12, 15, 25), use the first two or three digits together: 27 ÷ 12 = 2 r 3.
5. **Check by multiplying back:** 3.65 × 5 = 18.25 ✔. For money, write the answer with 2 decimal places and a $ sign.



Small red digits are carried remainders.

Short division (the bus stop): 18.25 ÷ 5 = 3.65.

Worksheet questions, worked out

WORKSHEET · EX 4.3 · Q4

In a supermarket, five chickens cost $18.25. What is the cost of one chicken?

Many → one, so divide: 18.25 ÷ 5 (estimate 20 ÷ 5 = 4)

18 ÷ 5 = 3 r 3; 32 ÷ 5 = 6 r 2; 25 ÷ 5 = 5

One chicken costs **$3.65** (check: 3.65 × 5 = 18.25 ✔)

WORKSHEET · EX 4.3 · Q5

Six friends have a meal in a restaurant. The total bill is $145.50. They share the bill equally. How much do they each pay?

> Share equally, so divide: 145.50 ÷ 6 (estimate 150 ÷ 6 = 25)
> 14 ÷ 6 = 2 r 2; 25 ÷ 6 = 4 r 1; 15 ÷ 6 = 2 r 3; 30 ÷ 6 = 5
> Each friend pays **$24.25** (check: 24.25 × 6 = 145.50 ✔)

WORKSHEET · EX 4.4 · Q3

Maggie pays $9.28 for 8 m of ribbon. What is the cost of the ribbon, per metre?

> Per metre means ÷ 8: 9 ÷ 8 = 1 r 1; 12 ÷ 8 = 1 r 4; 48 ÷ 8 = 6
> Cost per metre = **$1.16** (check: 1.16 × 8 = 9.28 ✔)

WORKSHEET · EX 4.4 · Q2 AND Q6

Q2: Work out. Q6: Complete these divisions.

Question	Bus-stop steps	Answer
Q2 a) 8.654 ÷ 2	8÷2 = 4; 6÷2 = 3; 5÷2 = 2 r 1; 14÷2 = 7	**4.327**
Q2 b) 8.922 ÷ 6	8÷6 = 1 r 2; 29÷6 = 4 r 5; 52÷6 = 8 r 4; 42÷6 = 7	**1.487**
Q2 c) 32.925 ÷ 5	32÷5 = 6 r 2; 29÷5 = 5 r 4; 42÷5 = 8 r 2; 25÷5 = 5	**6.585**
Q2 d) 58.912 ÷ 8	58÷8 = 7 r 2; 29÷8 = 3 r 5; 51÷8 = 6 r 3; 32÷8 = 4	**7.364**
Q6 a) 27.852 ÷ 12	27÷12 = 2 r 3; 38÷12 = 3 r 2; 25÷12 = 2 r 1; 12÷12 = 1	**2.321**
Q6 b) 46.875 ÷ 15	46÷15 = 3 r 1; 18÷15 = 1 r 3; 37÷15 = 2 r 7; 75÷15 = 5	**3.125**
Q6 c) 782.825 ÷ 25	78÷25 = 3 r 3; 32÷25 = 1 r 7; 78÷25 = 3 r 3; 32÷25 = 1 r 7; 75÷25 = 3	**31.313**

WATCH OUT!

Put the point in the answer first, straight above the point inside the bus stop. A point in the wrong place makes 3.65 into 36.5. Don't forget the remainders: each one is carried to the **next** digit. And for money, $39 is fine, but $3.6 should be written **$3.60**.

EXAM TIP

Words like **each**, **per**, **one** and **share equally** mean divide. Always check by multiplying back: 2.321 × 12 = 27.852 ✔.

✏️ **TRY THESE (ANSWERS AT THE BACK)**

a. Work out 9.636 ÷ 3.

b. Four cinema tickets cost $27.40. What is the cost of one ticket?

c. Work out 51.75 ÷ 15.

6 Collecting Data

Naming the type of data, using questionnaires, and finding the mode, median and mean of a set of numbers.

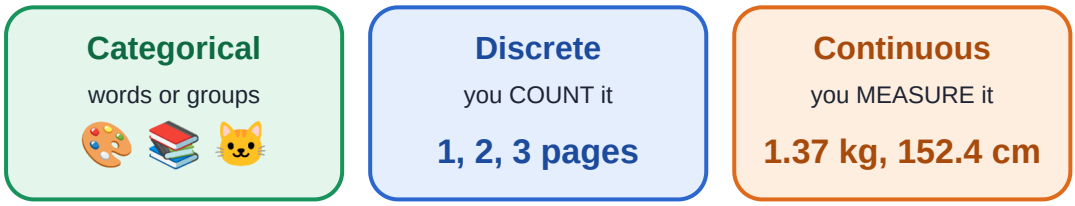
Big idea: Data is information we collect to answer a question. First decide **what kind** of data it is. Then use one number, an **average**, to describe the whole set: the **mode**, the **median** or the **mean**.

6.1 Types of data and questionnaires

Every piece of data is one of three types. Ask yourself: is it a **word**, a number you **count**, or a number you **measure**?

The method

1. Is the answer a **word or a group** (a colour, a subject, a pet)? It is **categorical**.
2. Is it a number you **count** or that can only be certain values (pages, people, shoe size 5, 5½, 6)? It is **discrete**.
3. Is it a number you **measure** that can be any value, even 1.37 (mass, height, time, length)? It is **continuous**.



The three types of data. Words → categorical; counting → discrete; measuring → continuous.

Worksheet questions, worked out

WORKSHEET · EX 6.1 · Q1 AND Q2

Q1 Choose the correct word (discrete, continuous, categorical) to describe the following. Q2 Here are some facts about a person. Write down the type of data for each fact.

Fact	Ask yourself	Type
Q1 1) The mass of a book	measured on scales, any value	**continuous**
Q1 2) The colour of a book	a word	**categorical**
Q1 3) The number of pages in a book	counted	**discrete**
Q2 a) age, in years	whole years: 11, 12, 13	**discrete**
Q2 b) shoe size	only certain sizes: 5, 5½, 6	**discrete**
Q2 c) height	measured, any value	**continuous**
Q2 d) time to travel to school	measured, any value	**continuous**
Q2 e) favourite subject	a word	**categorical**

WORKSHEET · GETTING STARTED · Q1 AND Q2

a) What is a questionnaire? b) Why would you use a questionnaire?

> a) **A set of questions used in a survey.**
>
> b) **To investigate a statistical question or to test a prediction.** (for example: "Do most students in my class walk to school?")

WATCH OUT!

A number is not always continuous. Age in years and shoe size are numbers, but they jump from one value to the next, so they are **discrete**. Continuous data is **measured** and can be any value in between.

MEMORY TRICK

Count → Discrete. Measure → Continuous. Words → Categorical. "You can't have 2.5 people, but you can be 152.5 cm tall."

✏️ **TRY THESE (ANSWERS AT THE BACK)**

a. What type of data is the number of goals scored in a football match?

b. What type of data is the length of a pencil?

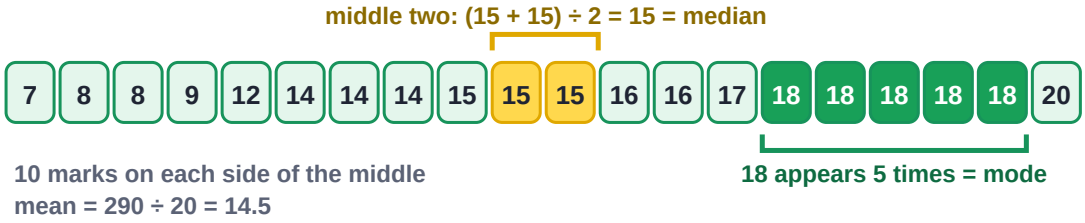
c. What type of data is a person's favourite fruit?

6.2 Mode, median and mean

An **average** is one number that sums up a whole set of data. There are three kinds, and each is found in its own way.

The method

1. **Mode**: the value that appears **most often**. Count how many times each value appears.
2. **Median**: the **middle** value. First **put the numbers in order**, smallest to largest. With an odd number of values, take the one in the middle. With an **even** number of values there are two in the middle: add them and divide by 2.
3. **Mean**: **add up** all the values, then **divide by how many** values there are.



The 20 test marks in order. The 10th and 11th marks (yellow) are in the middle; the 18s (green) are the most common.

Worksheet questions, worked out

WORKSHEET · GETTING STARTED · Q4

Here are 20 test marks: 7 8 8 9 12 14 14 14 15 15 15 16 16 17 18 18 18 18 18 20. a) Which mark is the mode? b) Which mark is the median? c) The sum of the marks is 290. What is the mean mark?

a) Count each mark: 14 appears 3 times, 15 appears 3 times, 18 appears **5 times**. Mode = **18**

b) The marks are already in order. There are 20 marks (even), so the middle two are the 10th and 11th: 15 and 15.

Median = (15 + 15) ÷ 2 = 30 ÷ 2 = **15**

c) Mean = sum ÷ how many = 290 ÷ 20 = **14.5** (the mean does not have to be one of the marks)

WATCH OUT!

Always sort the numbers before finding the median. The middle of a jumbled list is not the median. And for the **mean**, divide by **how many** numbers there are (20 here), not by the largest number.

MEMORY TRICK

MOde = MOst. MEDian = MIDDLE (like the median strip in the middle of a road). **MEAN** is the "mean" one: it makes you do the most work, add and divide.

EXAM TIP

To find where the middle is: for **n** numbers, the median is the **(n + 1) ÷ 2**th value. For 20 marks that is the 10.5th, so halfway between the 10th and 11th.

✏️ **TRY THESE (ANSWERS AT THE BACK)**

Use these numbers: **4, 9, 2, 7, 4, 10**

a. Find the mode.

b. Find the median.

c. Find the mean.

9 Sequences & Functions

Doing calculations in the right order, working with negative numbers, finding rules for sequences and patterns, the nth term, and function machines.

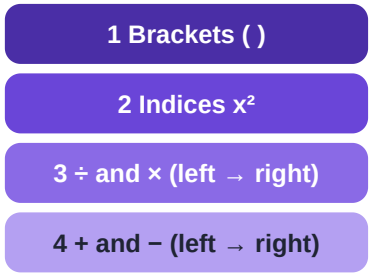
Big idea: Maths follows **rules**. BIDMAS tells you the order to work in. A sequence has a **rule** that makes each term. A function machine has a **rule** that turns each input into an output. **Find the rule, and you can work out anything.**

9.1 Order of operations & negative numbers

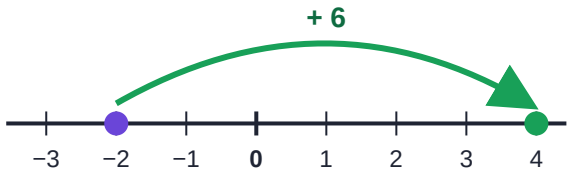
When a calculation has more than one operation, everyone must do them in the **same order**, or we would all get different answers. That order is called **BIDMAS**.

The method

1. **B**rackets first: work out anything inside ( ).
2. **I**ndices next: powers like $3^2 = 3 \times 3 = 9$.
3. **D**ivide and **M**ultiply next. They are **equal**, so do them from **left to right**.
4. **A**dd and **S**ubtract last. They are also **equal**, so do them from **left to right**.
5. A fraction line works like a bracket: $\frac{12 + 9}{3}$ means $(12 + 9) \div 3$.
6. Negatives: adding moves **right** on the number line, subtracting moves **left**. For × and ÷: **same signs give +, different signs give −**.



The BIDMAS ladder: start at the top. ÷ and × share a step, and so do + and −.



−2 + 6: start at −2 and jump 6 to the right. You land on 4.

Worksheet questions, worked out

WORKSHEET · CH 9 GETTING STARTED · Q1

Work out.

a) 8 + 6	b) 17 − 5	c) −2 + 6	d) 9 − 12
14	**12**	**4** *(−2, jump 6 right)*	**−3** *(9, jump 12 left)*
e) 5 × 7	**f) 24 ÷ 6**	**g) −3 × 6**	**h) 35 ÷ −7**
35	**4**	**−18** *(− × + = −)*	**−5** *(+ ÷ − = −)*

WORKSHEET · CH 9 GETTING STARTED · Q2

Complete these calculations.

Question	Think	Missing number
a) 4 + □ = 11	11 − 4	**7**
b) 23 − □ = 18	23 − 18	**5**
c) □ + 3 = 12	12 − 3	**9**

d) −3 + □ = 3	from −3 up to 3 is 6 steps	**6**
e) □ − 8 = −5	−5 + 8	**3**
f) 10 − □ = −4	from 10 down to −4 is 14 steps	**14**
g) 3 × □ = 12	12 ÷ 3	**4**
h) □ × 12 = 96	96 ÷ 12	**8**
i) □ ÷ −2 = □	many answers work, for example	**10 ÷ −2 = −5**

WORKSHEET · CH 9 GETTING STARTED · Q3

Use the correct order of operations.

Question	Working	Answer
a) 5 + 4 × 3	× first: 5 + 12	**17**
b) (5 + 4) × 3	brackets first: 9 × 3	**27**
c) 6 + 4 ÷ 2	÷ first: 6 + 2	**8**
d) (6 + 4) ÷ 2	brackets first: 10 ÷ 2	**5**
e) $\frac{12}{3} + 9$	÷ first: 12 ÷ 3 = 4, then 4 + 9	**13**
f) $\frac{12 + 9}{3}$	top first: 21 ÷ 3	**7**
g) −2 × 3 + 10	× first: −6 + 10	**4**
h) $-5 - \frac{10}{5}$	÷ first: 10 ÷ 5 = 2, then −5 − 2	**−7**

WATCH OUT!

Don't just work from left to right. 5 + 4 × 3 is not 9 × 3 = 27. The × comes first: 5 + 12 = 17. And watch the signs: −6 + 10 is **+4** (start at −6, go up 10), and −5 − 2 is **−7** (go further down).

MEMORY TRICK

BIDMAS = Brackets, Indices, Divide/Multiply, Add/Subtract. D and M are a **team**, and A and S are a **team**: inside a team, go left to right. For × and ÷ with negatives: **same signs, plus; different signs, minus.**

✏️ **TRY THESE (ANSWERS AT THE BACK)**

a. 20 − 12 ÷ 4

b. (−3 + 7) × 2

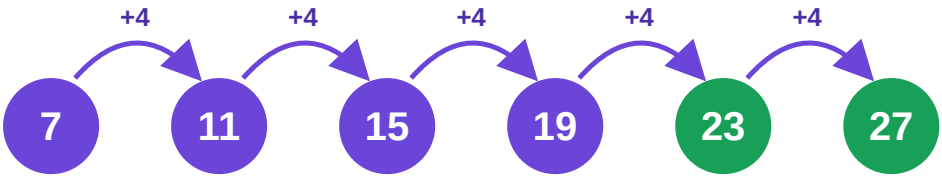
c. −4 × 5 + 6

9.2 Sequences & patterns

A **sequence** is a list of numbers that follow a rule. Each number is a **term**. The **term-to-term rule** tells you how to get from one term to the next.

The method

1. Look at the **gap** between each term and the next one.
2. If the gap is the same each time, the rule is **add** (going up) or **subtract** (going down) that number. Some rules multiply or divide instead.
3. Use the rule on the last term to get the **next terms**.
4. For missing terms in the middle, find the rule from two terms that are next to each other, then fill the gaps.
5. A sequence that ends is **finite**. A sequence with "…" at the end goes on for ever: it is **infinite**.



7, 11, 15, 19 … The gap is always 4, so the term-to-term rule is "add 4". The next two terms are 23 and 27.

Worksheet questions, worked out

WORKSHEET · EX 9.1 · Q1 AND Q2

Q1: Find the term-to-term rule and the next two terms. Q2: For each of these infinite sequences, write down (i) the term-to-term rule (ii) the next two terms.

Sequence	Gap	(i) Rule	(ii) Next two terms
Q1 a) 7, 11, 15, 19, …	11 − 7 = 4	**add 4**	**23, 27**
Q1 b) 40, 35, 30, 25, …	35 − 40 = −5	**subtract 5**	**20, 15**
Q2 a) 2, 4, 6, 8, …	+2	**add 2**	**10, 12**
Q2 b) 1, 4, 7, 10, …	+3	**add 3**	**13, 16**
Q2 c) 5, 9, 13, 17, …	+4	**add 4**	**21, 25**
Q2 d) 3, 8, 13, 18, …	+5	**add 5**	**23, 28**
Q2 e) 30, 28, 26, 24, …	−2	**subtract 2**	**22, 20**
Q2 f) 17, 14, 11, 8, …	−3	**subtract 3**	**5, 2**

WORKSHEET · EX 9.1 · Q3

Write down the first three terms of each of these sequences.

	First term	Term-to-term rule	First three terms
a)	1	add 5	**1, 6, 11**
b)	45	subtract 7	**45, 38, 31**
c)	6	multiply by 2	**6, 12, 24**
d)	60	divide by 2	**60, 30, 15**

WORKSHEET · EX 9.1 · Q4

Copy these finite sequences and fill in the missing terms.

Sequence	Rule	Complete sequence (missing terms in bold)
a) 2, 5, □, 11, □, 17, 20	add 3	2, 5, **8**, 11, **14**, 17, 20
b) 5, 11, 17, □, □, 35, □	add 6	5, 11, 17, **23**, **29**, 35, **41**
c) 26, 23, □, □, 14, □, 8	subtract 3	26, 23, **20**, **17**, 14, **11**, 8
d) 90, 82, □, 66, □, 50, □	subtract 8	90, 82, **74**, 66, **58**, 50, **42**
e) 8, □, □, 32, 40, □, □	add 8	8, **16**, **24**, 32, 40, **48**, **56**
f) □, □, 28, 23, □, □, 8	subtract 5	**38**, **33**, 28, 23, **18**, **13**, 8

In f), work backwards from 28: the rule is subtract 5, so going back you add 5: 28 + 5 = 33, 33 + 5 = 38.

WORKSHEET · EX 9.1 · Q5

Write down whether each of these sequences is finite or infinite.

a) 4, 6, 8, 10, … → ends with "…": **infinite**

b) 3, 5, 7, 9, 11, 13, 15 → it stops: **finite**

c) 85, 75, 65, 55, 45, 35 → it stops: **finite**

d) 100, 97, 94, 91, 88, … → ends with "…": **infinite**

WORKSHEET · EX 9.2 · Q1 AND Q2

Q1: This sequence of patterns is made from dots (4 dots, 7 dots, 10 dots). Q2: This sequence of patterns is made from squares (5 squares, 8 squares, 11 squares).

Pattern number	1	2	3	4	5	6	8
Q1 Dots	4	7	10	**13**	**16**		
Q2 Squares	5	8	11	**14**	**17**	**20**	**26**

Q1 a) Draw patterns 4 and 5 with **13** and **16** dots: add 3 more dots each time.

Q1 b) Sequence of numbers of dots: **4, 7, 10, 13, 16**

Q1 c) Term-to-term rule: **add 3**

Q1 d) How the sequence is formed: **start with 4 dots, then add 3 more dots to the previous pattern each time**

Q2 a), b) Patterns 4 and 5 have **14** and **17** squares. c) Rule: **add 3**

Q2 d) (i) Pattern 6: 17 + 3 = **20** (ii) Pattern 8: 20 + 3 + 3 = **26**

WATCH OUT!

Check the direction. If the terms get smaller, the rule is **subtract**, not add. And always check your rule works for **every** gap, not just the first one.

EXAM TIP

Write the rule in words, like "add 4" or "subtract 5". Writing just "4" does not say whether it goes up or down.

✏️ **TRY THESE (ANSWERS AT THE BACK)**

a. Find the term-to-term rule and the next two terms: 3, 10, 17, 24, …

b. The first term is 80 and the term-to-term rule is subtract 6. Write down the first three terms.

c. Fill in the missing terms: 4, □, 16, 22, □

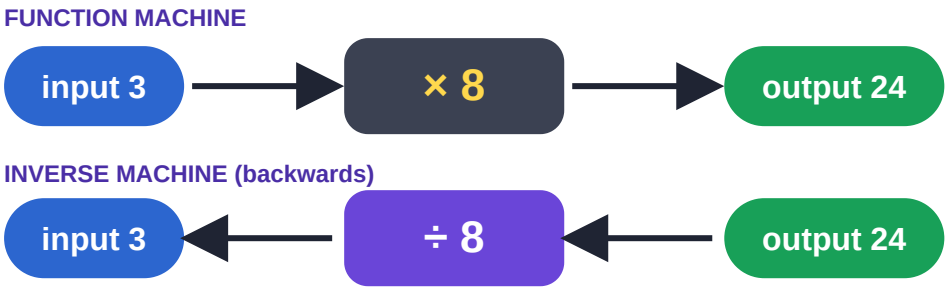
9.3 The nth term & function machines

The **nth term rule** links each term to its **position number** n (1st, 2nd, 3rd …). With it you can jump straight to any term, like the 20th, without listing them all. A **function machine** uses a rule to turn an **input** into an **output**.

The method

1. Make a table: **position number** 1, 2, 3 … on top, **term** underneath.
2. If the terms are the **times table** of a number (6, 12, 18 …), the rule is "term = 6 × position", so the nth term is **6n**.
3. If the terms go up by 1 (10, 11, 12 …), compare each term with its position: 10 is 9 more than 1, so the nth term is **n + 9**.
4. To find a term, put its position in for n: the 10th term of 6n is 6 × 10 = 60.
5. Function machine: do the rule to each **input** to get the **output**. To go **backwards** from an output, use the **inverse** (opposite): + ↔ −, × ↔ ÷.

Position number (n)	1	2	3	4	5	n
Working	1 × 6	2 × 6	3 × 6	4 × 6	5 × 6	n × 6
Term	6	12	18	24	30	**6n**



Forwards: 3 × 8 = 24. Backwards with the inverse: 24 ÷ 8 = 3.

Worksheet questions, worked out

WORKSHEET · EX 9.3 · Q1

a) Complete the working shown to work out the nth term rule for this sequence: 6, 12, 18, 24, 30, … b) Use your rule to work out the (i) 10th term (ii) 15th term.

> a) Every term is **6 ×** its position number (see the table above), so **nth term = 6n**
> b) (i) 10th term = 6 × 10 = **60** (ii) 15th term = 6 × 15 = **90**

WORKSHEET · EX 9.3 · Q2 AND Q3

Q2: For each of these sequences, work out the nth term rule. Q3: For each of the sequences in Q2, use its nth term rule to work out the (i) eighth term (ii) 20th term.

Sequence	nth term	(i) 8th term	(ii) 20th term
a) 5, 10, 15, 20, 25, …	**5n**	5 × 8 = **40**	5 × 20 = **100**
b) 8, 16, 24, 32, 40, …	**8n**	8 × 8 = **64**	8 × 20 = **160**
c) 15, 30, 45, 60, 75, …	**15n**	15 × 8 = **120**	15 × 20 = **300**

WORKSHEET · EX 9.3 · Q5

For each of these sequences, work out its nth term rule.

Sequence	Compare term with position	nth term
a) 10, 11, 12, 13, 14, …	1st term 10 = 1 + 9	**n + 9**
b) 5, 6, 7, 8, 9, …	1st term 5 = 1 + 4	**n + 4**
c) 22, 23, 24, 25, 26, …	1st term 22 = 1 + 21	**n + 21**
d) 43, 44, 45, 46, 47, …	1st term 43 = 1 + 42	**n + 42**

WORKSHEET · EX 9.4 · Q1 AND Q2

Copy these function machines and work out the missing outputs. In Q2, use inverse operations to work out the missing inputs.

Machine	Inputs	Outputs
Q1 a) + 5	2, 3, 4	**7, 8, 9**
Q1 b) − 3	12, 8, 6	**9, 5, 3**
Q1 c) × 6	1, 3, 5	**6, 18, 30**
Q1 d) ÷ 4	12, 20, 40	**3, 5, 10**
Q2 a) + 7	2, 5, 9	**9, 12, 16**
Q2 b) − 5 *(inverse + 5)*	**7, 10, 15**	2, 5, 10

Q2 c) × 3 *(inverse ÷ 3)*	**4, 5, 10**	12, 15, 30
Q2 d) ÷ 2	8, 12, 18	**4, 6, 9**

WORKSHEET · EX 9.4 · Q4

Copy and complete these inverse function machines and tables of values.

Machine	Inverse machine	Input	Output
a) input → − 13 → output	**input ← + 13 ← output**	20, 25, 51	7, 12, 38
b) input → × 8 → output	**input ← ÷ 8 ← output**	3, 7, 15	24, 56, 120
c) input → + 9 → output	**input ← − 9 ← output**	8, 15, 36	17, 24, 45

Check one in each row: 51 − 13 = 38 and 38 + 13 = 51; 15 × 8 = 120 and 120 ÷ 8 = 15; 36 + 9 = 45 and 45 − 9 = 36.

WATCH OUT!

The term-to-term rule is not the nth term. For 5, 10, 15 … the term-to-term rule is "add 5", but the nth term is **5n**, not n + 5. Test it: n + 5 gives 1 + 5 = 6 for the 1st term, which is wrong. 5n gives 5 × 1 = 5. Correct!

TRY A NUMBER

Always check your nth term with n = 1 and n = 2. For n + 9: 1 + 9 = 10 and 2 + 9 = 11. Those match the first two terms, so the rule is right.

✏️ **TRY THESE (ANSWERS AT THE BACK)**

a. Find the nth term of 9, 18, 27, 36, … and use it to work out the 12th term.

b. Find the nth term of 14, 15, 16, 17, …

c. A function machine is × 7. The outputs are 42 and 63. Use the inverse to find the inputs.

10 Percentages

Changing between percentages, fractions and decimals, and finding fractions and percentages of amounts.

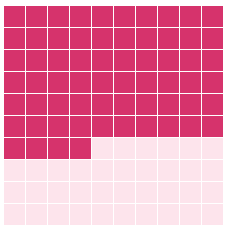
Big idea: Per cent means "out of 100". 64% is 64 out of every 100, which is $\frac{64}{100}$ or 0.64. A percentage, a fraction and a decimal are three ways of writing the **same amount**.

10.1 Percentages, fractions and decimals

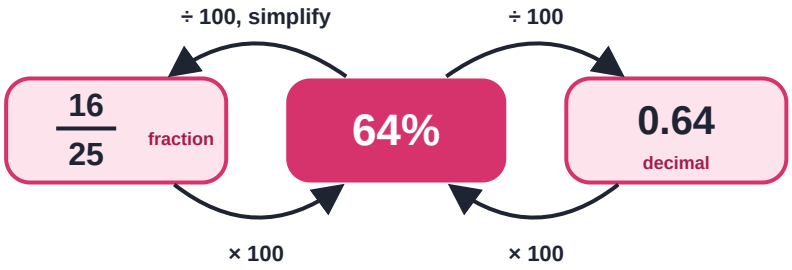
To change between the three forms, you only need two moves: **÷ 100** to leave a percentage, and **× 100** to make one.

The method

1. **Percentage → fraction:** write the number over 100, then **simplify**: divide the top and bottom by the same number until you can't any more. If the percentage has a decimal point (7.5%), multiply the top and bottom by 10 first: $\frac{7.5}{100} = \frac{75}{1000}$.
2. **Percentage → decimal:** divide by 100. The digits move **two places** to the right of the point: 40% = 0.40 = 0.4, 4% = 0.04.
3. **Fraction → percentage:** multiply by 100%. Turn the fraction into a decimal first (top ÷ bottom), then × 100: $\frac{1}{4}$ = 0.25, and 0.25 × 100 = 25%.
4. **Mixed number → percentage:** the whole number 1 is 100%. So $1\frac{1}{2}$ = 1.5 = 150%. Percentages over 100% mean "more than the whole".



64 out of 100 = 64%



64 squares out of 100 are shaded: that is 64%. Leaving a percentage: ÷ 100. Making a percentage: × 100.

Worksheet questions, worked out

WORKSHEET · EX 10.1 · Q2

Write these percentages as fractions. Write your answers in the simplest form.

Percentage	Over 100	Simplify	Answer
a) 60%	$\frac{60}{100}$	÷ 20 top and bottom	$\frac{3}{5}$
b) 61%	$\frac{61}{100}$	nothing divides both	$\frac{61}{100}$
c) 62%	$\frac{62}{100}$	÷ 2 top and bottom	$\frac{31}{50}$
d) 64%	$\frac{64}{100}$	÷ 2 → $\frac{32}{50}$, ÷ 2 again → $\frac{16}{25}$	$\frac{16}{25}$
e) 65%	$\frac{65}{100}$	÷ 5 top and bottom	$\frac{13}{20}$
f) 70%	$\frac{70}{100}$	÷ 10 top and bottom	$\frac{7}{10}$

WORKSHEET · EX 10.1 · Q4

Write these percentages as a decimal and as a fraction.

Percentage	Decimal (÷ 100)	Fraction (over 100, simplify)
a) 40%	**0.4**	$\frac{40}{100} \div 20 \rightarrow \mathbf{\frac{2}{5}}$
b) 4%	**0.04**	$\frac{4}{100} \div 4 \rightarrow \mathbf{\frac{1}{25}}$
c) 9%	**0.09**	already simplest: $\mathbf{\frac{9}{100}}$
d) 90%	**0.9**	$\frac{90}{100} \div 10 \rightarrow \mathbf{\frac{9}{10}}$
e) 5%	**0.05**	$\frac{5}{100} \div 5 \rightarrow \mathbf{\frac{1}{20}}$

WORKSHEET · EX 10.1 · Q5

Write these fractions as percentages.

Fraction	Decimal (top ÷ bottom)	× 100
a) $\frac{1}{4}$	0.25	**25%**
b) $\frac{2}{5}$	0.4	**40%**
c) $\frac{4}{5}$	0.8	**80%**
d) $\frac{7}{50}$	0.14	**14%**
e) $\frac{7}{20}$	0.35	**35%**
f) $\frac{7}{25}$	0.28	**28%**

WORKSHEET · EX 10.2 · Q2

Write these percentages as decimals and as fractions. Write each in its simplest form.

Percentage	Decimal (÷ 100)	Fraction (× 10 top and bottom, then simplify)
a) 7.5%	**0.075**	$\frac{7.5}{100} = \frac{75}{1000}, \div 25 \rightarrow \mathbf{\frac{3}{40}}$
b) 62.5%	**0.625**	$\frac{62.5}{100} = \frac{625}{1000}, \div 125 \rightarrow \mathbf{\frac{5}{8}}$
c) 1.5%	**0.015**	$\frac{1.5}{100} = \frac{15}{1000}, \div 5 \rightarrow \mathbf{\frac{3}{200}}$
d) 47.5%	**0.475**	$\frac{47.5}{100} = \frac{475}{1000}, \div 25 \rightarrow \mathbf{\frac{19}{40}}$
e) 32.5%	**0.325**	$\frac{32.5}{100} = \frac{325}{1000}, \div 25 \rightarrow \mathbf{\frac{13}{40}}$

> Simplifying in small steps is fine too: $\frac{625}{1000} \div 5 \rightarrow \frac{125}{200} \div 5 \rightarrow \frac{25}{40} \div 5 \rightarrow \frac{5}{8}$. Keep going until nothing divides both.

WORKSHEET · EX 10.2 · Q9

Write each of these mixed numbers as a percentage.

Mixed number	Improper fraction	Decimal	× 100
a) $1\frac{1}{2}$	$\frac{3}{2}$	1.5	**150%**
b) $1\frac{1}{4}$	$\frac{5}{4}$	1.25	**125%**
c) $1\frac{3}{4}$	$\frac{7}{4}$	1.75	**175%**
d) $1\frac{3}{10}$	$\frac{13}{10}$	1.3	**130%**

e) $1\frac{7}{10}$	$\frac{17}{10}$	1.7	**170%**
f) $2\frac{1}{4}$	$\frac{9}{4}$	2.25	**225%**

WATCH OUT!

7.5% is 0.075, not 0.75 and not 7.5. Dividing by 100 moves the digits **two** places, so put in a zero: 7.5 → 0.75 (÷ 10) → 0.075 (÷ 100). Check: 7.5% is less than 10% = 0.1, and 0.075 is less than 0.1.

MEMORY TRICK

Leaving % → ÷ 100. Making % → × 100. "Per cent" is like "century": 100 years. Every percentage is secretly "out of 100".

✏️ **TRY THESE (ANSWERS AT THE BACK)**

a. Write 35% as a fraction in its simplest form.

b. Write 12.5% as a decimal and as a fraction in its simplest form.

c. Write $1\frac{2}{5}$ as a percentage.

10.2 Percentages of amounts

"Of" means **multiply**. To find a fraction or a percentage of an amount, cut the amount into equal parts, then take the parts you need.

The method

1. **Fraction of an amount:** divide by the **bottom**, then multiply by the **top**. $\frac{3}{4}$ of 60 g: 60 ÷ 4 = 15, then 15 × 3 = 45 g.
2. **10%** of an amount: divide by 10. **1%**: divide by 100 (or divide the 10% by 10).
3. **0.5%** is half of 1%, and **0.1%** is 1% ÷ 10.
4. Build the percentage you want by **adding chunks**: 3.5% = 1% + 1% + 1% + 0.5%.



600 kg is 100%. Cut it into ten 10% chunks of 60 kg, then cut one chunk into ten 1% pieces of 6 kg.

Worksheet questions, worked out

WORKSHEET · EX 10.1 · Q9

Work out: a) $\frac{1}{2}$ of 60 grams, b) $\frac{3}{4}$ of 60 grams, c) $\frac{4}{5}$ of 60 grams, d) $\frac{7}{10}$ of 60 grams, e) $\frac{11}{20}$ of 60 grams.

Fraction of 60 g	÷ bottom	× top	Answer
a) $\frac{1}{2}$	60 ÷ 2 = 30	30 × 1 = 30	**30 g**
b) $\frac{3}{4}$	60 ÷ 4 = 15	15 × 3 = 45	**45 g**
c) $\frac{4}{5}$	60 ÷ 5 = 12	12 × 4 = 48	**48 g**
d) $\frac{7}{10}$	60 ÷ 10 = 6	6 × 7 = 42	**42 g**
e) $\frac{11}{20}$	60 ÷ 20 = 3	3 × 11 = 33	**33 g**

WORKSHEET · EX 10.2 · Q5

Work out: (i) 1% of 600 kg, (ii) 1.5% of 600 kg, (iii) 3.5% of 600 kg, (iv) 2.2% of 600 kg.

1% of 600 kg = 600 ÷ 100 = 6 kg. Every answer is built from this.

(i) 1% = **6 kg**

(ii) 1.5% = 1% + 0.5% = 6 + 3 = **9 kg** (or 1.5 × 6 = 9)

(iii) 3.5% = 3% + 0.5% = 18 + 3 = **21 kg** (or 3.5 × 6 = 21)

(iv) 2.2% = 2% + 0.2% = 12 + 1.2 = **13.2 kg** (0.1% = 0.6 kg, so 0.2% = 1.2 kg)

With fractions: 2.2% = $\frac{22}{1000}$, and $\frac{22}{1000} \times 600 = \frac{22 \times 3}{5} = \frac{66}{5}$ = 13.2 kg. Same answer.

WATCH OUT!

1% means ÷ 100, not ÷ 10. 1% of 600 kg is 6 kg. Dividing by 10 gives 60 kg, which is **10%**. Check your answer makes sense: 1.5% of 600 kg must be a small part of it.

TRY A NUMBER

Check with an easy one: 50% is a half, and half of 600 kg is 300 kg. Using chunks: 50% = five 10% chunks = 5 × 60 = 300 kg. It matches.

✏️ **TRY THESE (ANSWERS AT THE BACK)**

a. Work out $\frac{3}{5}$ of 80 grams.

b. Work out 15% of 600 kg.

c. Work out 2.5% of 400 kg.

🤖 Maths Cheat Sheet

Words → algebra

- **more than, plus, increased by** → +
- **less than, take away, decreased by** → −
- **times, multiplied by, twice** → ×
- **divided by, shared, half** → ÷
- 3 × y = **3y** · p × q = **pq** · y ÷ 2 = $\frac{y}{2}$
- **x less than y = y − x** (start with y)
- subtract **from 52** → 52 − …

Substitute & simplify

- Swap each letter for its number, then use BIDMAS: 2m + n, m = 7, n = 6 → 14 + 6 = **20**
- ab means a × b; 2h means 2 × h
- **Like terms** have the same letter: 5x + 4x = 9x · x on its own = 1x
- Collect each kind: 2g + 8 + 3g − 5 = **5g + 3**
- Pyramid: each block = the two below added

Brackets & equations

- **Expand:** multiply **every** term inside: 2(3b − 4) = **6b − 8**
- **Solve:** do the **inverse** to **both sides**: + ↔ −, × ↔ ÷
- x − 18 = 30 → x = 30 + 18 = **48** · 12x = 72 → x = **6**
- **Check** by putting the answer back in

Inequalities

- **<** less than · **>** greater than: the small end points at the smaller number
- x < 3: **open circle** at 3, arrow **left**
- y > −7: smallest integer is **−6**

Decimals: order & mental

- Line up the points; fill gaps with zeros: 2.19 = 2.190 < 2.205
- Compare digit by digit from the left
- Round, then adjust: 4.3 + 7.9 = 4.3 + 8 − 0.1 = **12.2**
- 3 × 0.004 = **0.012** (3 × 4 = 12, three decimal places)

× and ÷ decimals

- **×:** ignore the points, multiply, count the decimal places, put them back: 3.2 × 52 → 32 × 52 = 1664 → **166.4**
- **÷:** bus stop, point goes straight up: 18.25 ÷ 5 = **3.65**
- **Estimate first**, check by multiplying back
- Money: 2 decimal places ($3.60)

Data

- **Count** → discrete · **Measure** → continuous · **Words** → categorical
- **Mode** = most common
- **Median** = middle, **after sorting** (two middles: halfway)
- **Mean** = total ÷ how many: 290 ÷ 20 = **14.5**

BIDMAS & negatives

- **B**rackets → **I**ndices → **D M** (equal, left to right) → **A S** (equal, left to right)
- A fraction line means ÷: $\frac{12}{3}$ + 9 = 4 + 9 = **13**
- −2 + 6 = **4** (move right) · 9 − 12 = −3
- × ÷: same signs → +, different signs → −: −2 × 3 = −6

Sequences & machines

- **Term-to-term rule:** what you do to get the next term (add 4)
- **nth term:** position → term. Jumps of 5 starting at 5 → **5n** (not n + 5)
- 5n: 8th term = 40, 20th = 100
- Jumps of 1: n + 21 → 22, 23, 24…
- **Inverse machine:** undo each step, in reverse order

Percentages

- **Per cent = out of 100**
- % → decimal: **÷ 100** · decimal → %: **× 100**
- % → fraction: over 100, then simplify: 64% = $\frac{64}{100}$ = $\frac{16}{25}$
- 7.5% = **0.075** = $\frac{3}{40}$
- $1\frac{1}{2}$ = 150% (more than one whole)

% of an amount

- 10% = ÷ 10 · 1% = ÷ 100 · 50% = half · 25% = quarter
- Build it: 2.2% of 600 kg = 2 × 6 + 0.2 × 6 = **13.2 kg**
- $\frac{3}{4}$ of 60 g: ÷ 4, × 3 = **45 g**

Stuck? 🤖

- **Try a real number first**, then do the same with the letter
- **Estimate** before you calculate
- **Check**: put your answer back in, or work backwards

✔ Answer key: Try these

Check your answers. If one is wrong, read the working, then look back at that section.

Chapter 2

2.1 a) k − 6 *(start with k, take 6 away)*

2.1 b) 7n + 2

2.1 c) 20 − 3w *(subtract from 20, so 20 comes first)*

2.2 a) 21 *(4 × 6 − 3 = 24 − 3)*

2.2 b) 9 *((14 + 13) ÷ 3 = 27 ÷ 3)*

2.2 c) $128 *(P = M + E = 90 + 38)*

2.3 a) 5m *(7 + 2 − 4 = 5)*

2.3 b) 8a + 2 *(6a + 2a = 8a, 5 − 3 = 2)*

2.3 c) middle 6y and 7y, top 13y *(2y + 4y, 4y + 3y, 6y + 7y)*

2.4 a) 5k − 15 *(5 × k, 5 × 3, keep the −)*

2.4 b) 8x + 28 *(4 × 2x, 4 × 7)*

2.4 c) 18 − 6y *(3 × 6, 3 × 2y, keep the −)*

2.5 a) x = 14 *(23 − 9; check 14 + 9 = 23)*

2.5 b) x = 22 *(15 + 7; check 22 − 7 = 15)*

2.5 c) x = 7 *(56 ÷ 8; check 8 × 7 = 56)*

2.6 a) m is less than 3

2.6 b) k > −3

2.6 c) smallest integer 7; integer values 7, 8, 9, … *(6.5 is not whole; next whole number up is 7)*

Chapter 4

4.1 a) 0.75 *(0.80 vs 0.75: tenths 7 < 8)*

4.1 b) 5.306 < 5.31 *(5.306 vs 5.310: hundredths 0 < 1)*

4.1 c) 3.04, 3.399, 3.4, 3.41 *(pad: 3.040, 3.399, 3.400, 3.410)*

4.2 a) 5.5 *(9 − 3 = 6; 0.2 − 0.7 = −0.5; 6 − 0.5)*

4.2 b) 8.5 *(5.6 + 3 = 8.6, then − 0.1)*

4.2 c) 0.21 *(7 × 3 hundredths = 21 hundredths)*

4.3 a) 88.4 *(26 × 34 = 884, 1 decimal place; check 3 × 30 = 90)*

4.3 b) 364 *(7 × 52 = 350 + 14)*

4.3 c) 47.7 *(53 − 5.3)*

4.4 a) 3.212 *(9÷3 = 3; 6÷3 = 2; 3÷3 = 1; 6÷3 = 2)*

4.4 b) $6.85 *(27.40 ÷ 4: 27÷4 = 6 r 3; 34÷4 = 8 r 2; 20÷4 = 5)*

4.4 c) 3.45 *(51÷15 = 3 r 6; 67÷15 = 4 r 7; 75÷15 = 5)*

Chapter 6

6.1 a) discrete *(goals are counted: 0, 1, 2…)*

6.1 b) continuous *(length is measured and can be any value)*

6.1 c) categorical *(the answer is a word, like "mango")*

6.2 a) 4 *(4 appears twice; every other number once)*

6.2 b) 5.5 *(in order: 2, 4, 4, 7, 9, 10; middle two (4 + 7) ÷ 2 = 5.5)*

6.2 c) 6 *(sum 36 ÷ 6 numbers = 6)*

Chapter 9

9.1 a) 17 *(÷ first: 12 ÷ 4 = 3, then 20 − 3)*

9.1 b) 8 *(brackets first: −3 + 7 = 4, then 4 × 2)*

9.1 c) −14 *(× first: −4 × 5 = −20, then −20 + 6)*

9.2 a) add 7; 31, 38 *(24 + 7 = 31, 31 + 7 = 38)*

9.2 b) 80, 74, 68 *(80 − 6 = 74, 74 − 6 = 68)*

9.2 c) 10 and 28 *(the rule is add 6: 4, 10, 16, 22, 28)*

9.3 a) 9n; 12th term = 108 *(9 × 12)*

9.3 b) n + 13 *(1st term 14 = 1 + 13)*

9.3 c) 6 and 9 *(inverse ÷ 7: 42 ÷ 7 = 6, 63 ÷ 7 = 9)*

Chapter 10

10.1 a) $\frac{7}{20}$ *($\frac{35}{100}$, ÷ 5 top and bottom)*

10.1 b) 0.125 and $\frac{1}{8}$ *(12.5 ÷ 100 = 0.125; $\frac{125}{1000}$, ÷ 125)*

10.1 c) 140% *($1\frac{2}{5}$ = 1.4, × 100)*

10.2 a) 48 g *(80 ÷ 5 = 16, 16 × 3 = 48)*

10.2 b) 90 kg *(10% = 60 kg, 5% = 30 kg, 60 + 30 = 90)*

10.2 c) 10 kg *(1% = 4 kg, so 2% = 8 kg and 0.5% = 2 kg; 8 + 2 = 10)*